Ex8: Interactive Dashboards with Tableau

AIM: To design and build an interactive data visualization dashboard in Tableau using the Sample – Superstore dataset by creating bar, line, and scatter plot worksheets, organizing them on a dashboard, and configuring interactive filters and actions.

ALGORITHM:

1. Open Tableau and load the Sample – Superstore dataset from Kaggle, then explore the underlying data fields.
2. Create **Worksheet 1 – Bar Chart (Bar_Sales_by_Category)**:
    - Drag Category to Columns and SUM(Sales) to Rows.
    - Sort the bars in descending order.
    - Drag Region to Color and turn on "Show Mark Labels".
    - (Optional) Drag Region to Filters and select "Show Filter".
3. Create **Worksheet 2 – Line Chart (Line_Sales_Trend)**:
    - Drag Order Date to Columns and set it to continuous Month.
    - Drag SUM(Sales) to Rows and Region to Color.
    - Format the axis as Currency.
4. Create **Worksheet 3 – Scatter Plot (Scatter_Profit_vs_Sales)**:
    - Drag SUM(Sales) to Columns and SUM(Profit) to Rows.
    - Drag Sub-Category to Detail and Category to Color.
    - (Optional) Drag Segment to Shape and add reference lines for average values on both axes.
5. **Build the Dashboard (Sales Dashboard)**:
    - Create a new dashboard and set dimensions (e.g., 1200×800).
    - Add the title "Sales and Profit Dashboard" and arrange worksheets using layout containers: Line chart at top, Bar chart at bottom-left, and Scatter plot at bottom-right.
6. **Add Interactivity**:
    - Display filters for fields such as Region, Category, or Year.

- Configure Dashboard Filter Actions using the Bar chart as the source sheet and the Line and Scatter plots as target sheets triggered on selection.
- (Optional) Add Highlight Actions across charts.

7. **Publish/Export**: Save the finished interactive dashboard to Tableau Public or export as an Image/PDF.



RESULT: An interactive Tableau dashboard titled "Sales and Profit Dashboard" was successfully created using the Sample – Superstore dataset. The integrated worksheets, dynamic filters, and selection actions allow seamless interactive analysis of sales trends, regional performance, and product profitability